

CSE 311:

Data Communication

Instructor:

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Signal Characteristics: *digital source coding and error correction coding*

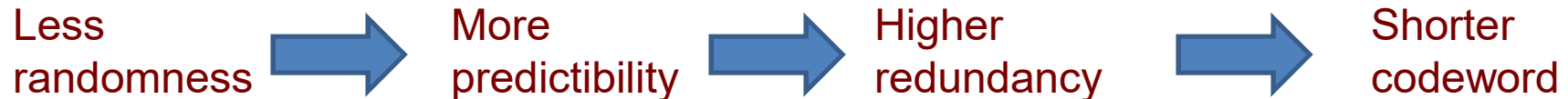
- 2 opposite procedures
- Digital source coding
 - An aggressive measure to **reduce redundancy**
- Error correction coding
 - A systematic **addition of redundancy** to detect/correct errors

Digital source Coding

- Removes redundancy
- Uses bandwidth as little as possible
- Related to randomness/predictability in data

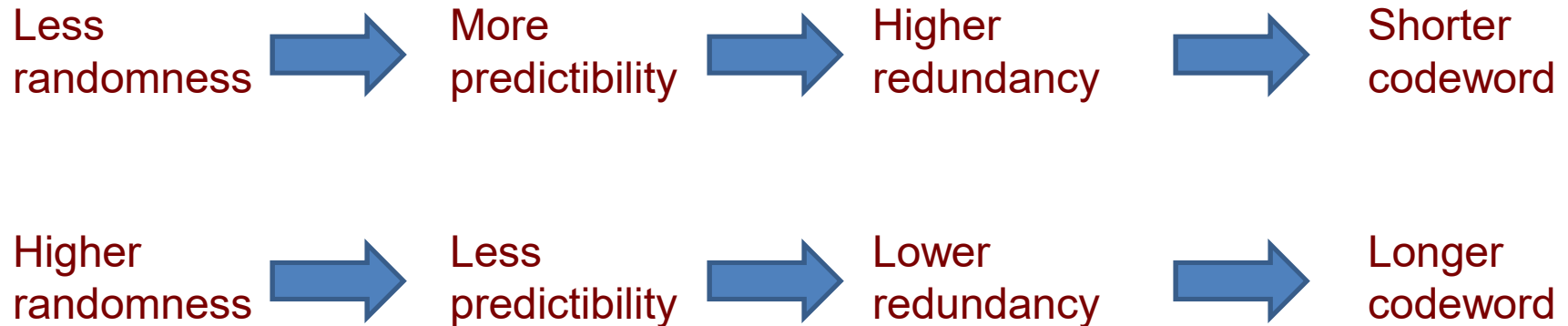
Digital source Coding

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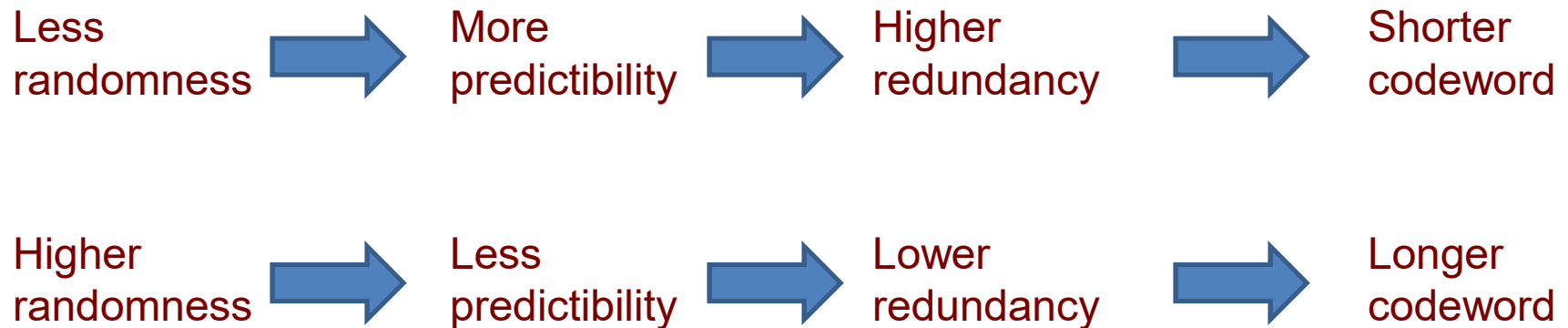
Digital source Coding

- Removes redundancy
- Uses bandwidth as little as possible
- Related to randomness/predictability in data

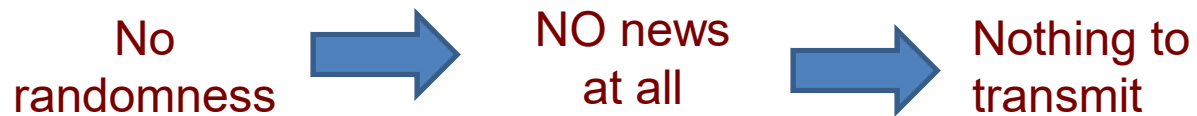


Digital source Coding

- Example: Morse code
 - Frequently occurring letters e, t, a: shorter codes
 - Less frequent letters: longer codes

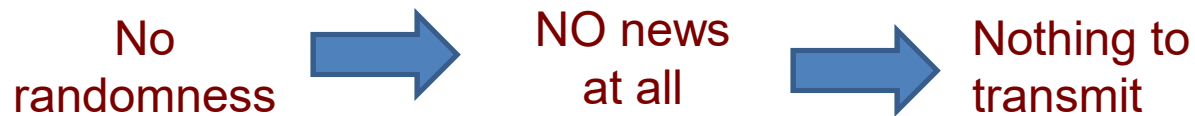


Digital source Coding



- A message with probability p ,
its *randomness or entropy* $= \log (1/p) = -\log (p)$

Digital source Coding



- A message with probability p ,
its *randomness or entropy* $= \log (1/p) = -\log (p)$

What happens when $p = 0$ and $p = 1$?

Error correction coding

- Add systemic redundancy
- 50% of English text is redundant
- Difficult to recover if error occurs in reduced text

Error correction coding

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- 50% of English text is redundant
- Difficult to recover if error occurs
- Error Detection: adding parity bit

Error correction coding

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Assume to transmit **0001**

Error correction coding

- Add systemic redundancy
- 50% of English text is redundant
- Difficult to recover if error occurs
- Error Detection: adding parity bit

0001**1**



Add this bit to make
even no. of 1's

Error correction coding

- Add systemic redundancy
- 50% of English text is redundant
- Difficult to recover if error occurs
- Error Detection: adding parity bit

0001**1**



Add this bit to make
even no. of 1's

Detects single bit error; however
cannot detect even no. of errors or
cannot locate or correct single bit
error

Signals and systems

Topics

- Definition, classification and properties of signals
- Examples of some useful signals
- Fourier Series of Periodic Signals

Classification of Signals

Review

- Based on **continuity in time axis**
 - Continuous time
 - Discrete time
- Based on **continuity in amplitude axis**
 - Continuous amplitude (**analog**)
 - Discrete amplitude (**digital**)

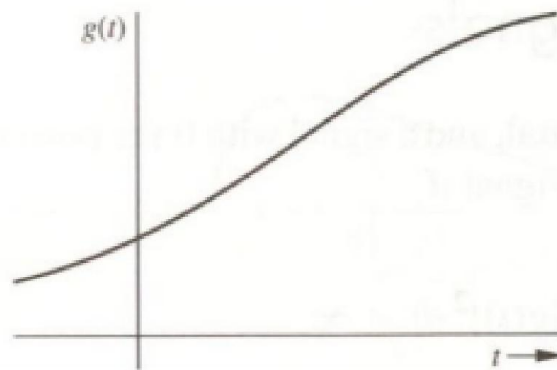
Classification of Signals

Review

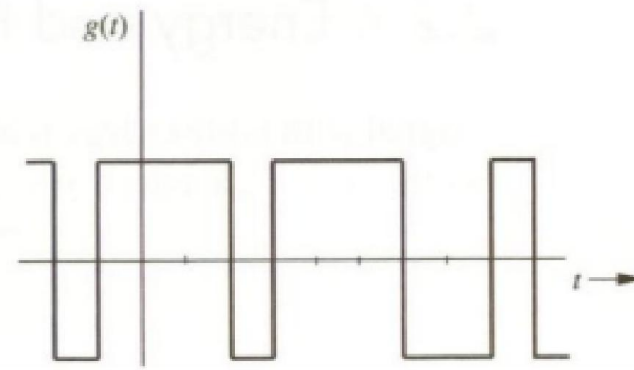
Continuous amplitude
(analog)

Discrete amplitude
(digital)

Continuous
time

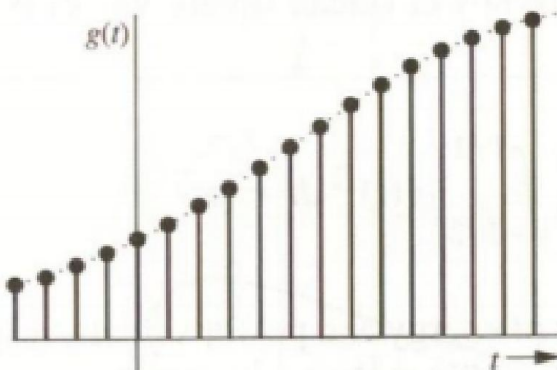


(a)

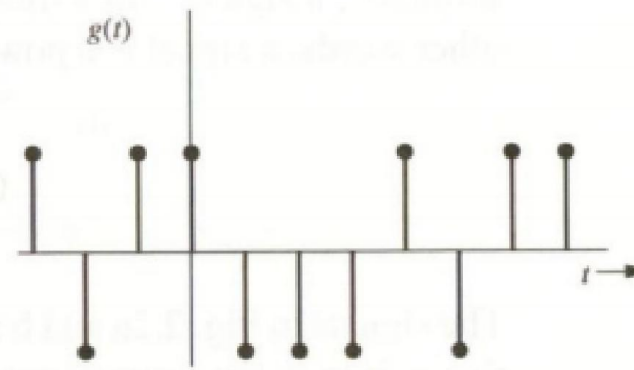


(b)

Discrete time



(c)



(d)

Periodic and Aperiodic signal

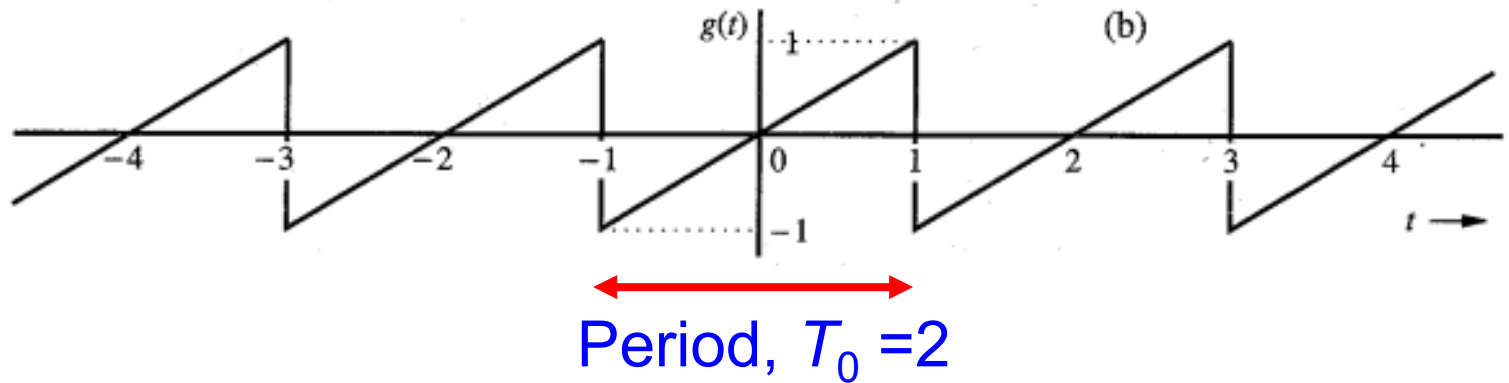
- Signal $g(t)$ is periodic for a positive constant T_0 so that

$$g(t) = g(t + T_0) \text{ for all } t$$

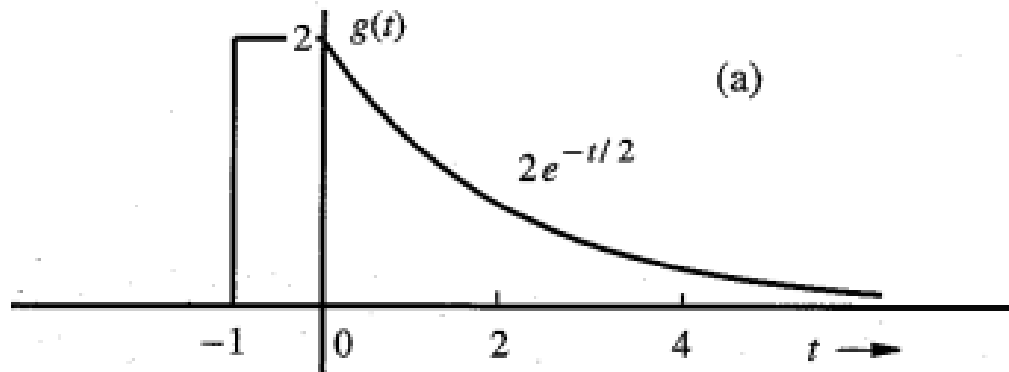
- Smallest T_0 is its period
- A signal is aperiodic if NOT periodic

Periodic and Aperiodic signal

Periodic
signal



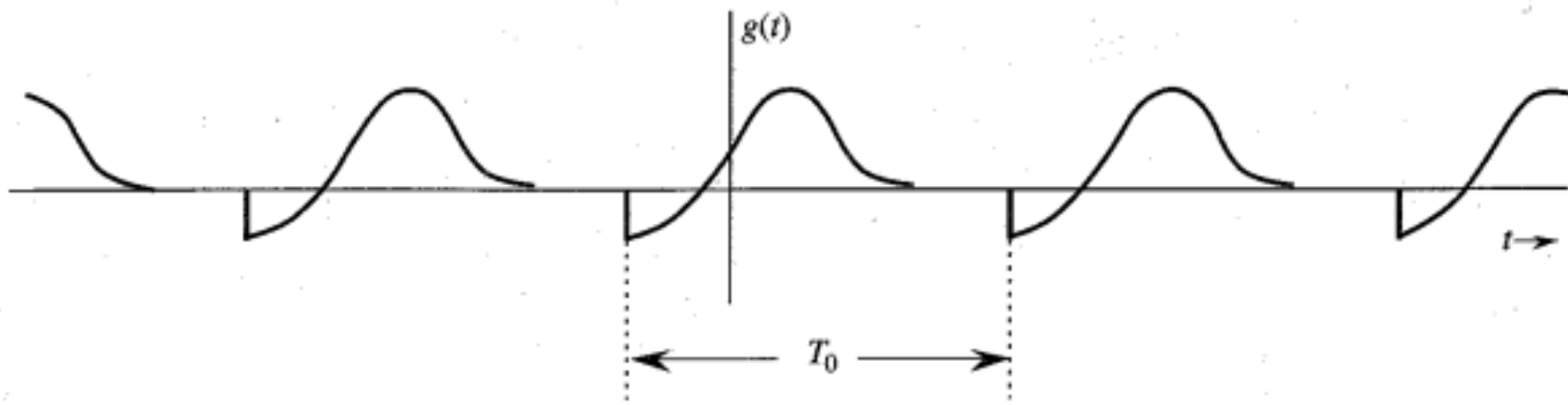
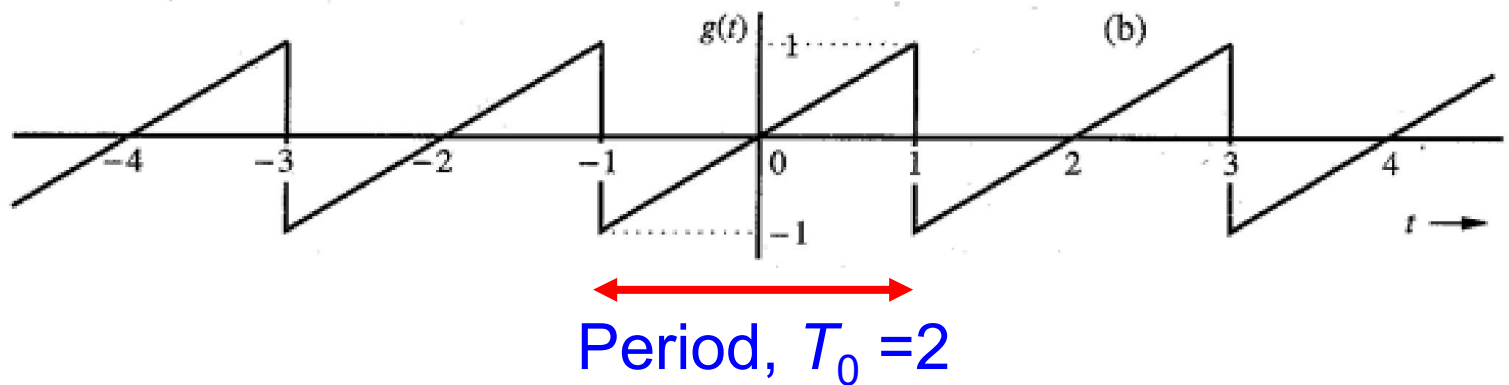
Aperiodic
signal



Periodic and Aperiodic signal

- Periodic signal starts at $t = -\alpha$ and continues forever

Periodic
signal

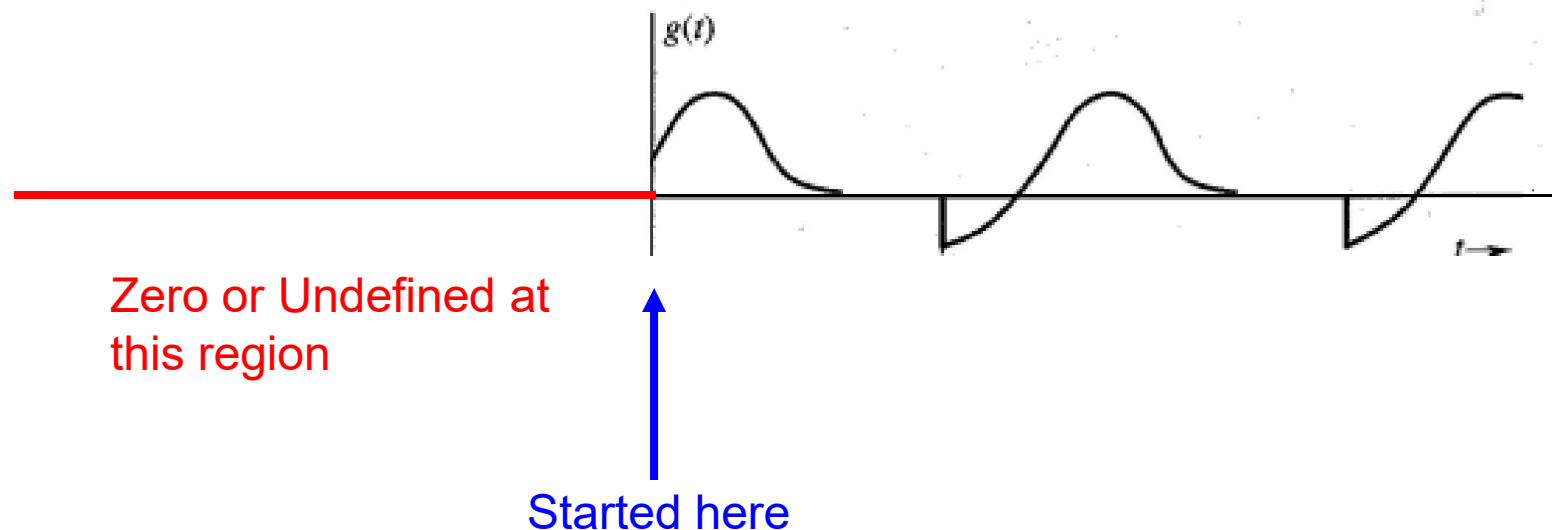


Periodic and Aperiodic signal

- Periodic signal starts at $t = -\alpha$ and continues forever
- cannot start at an finite time, say, $t = 0$, otherwise $g(t) = g(t + T_0)$ cannot be satisfied

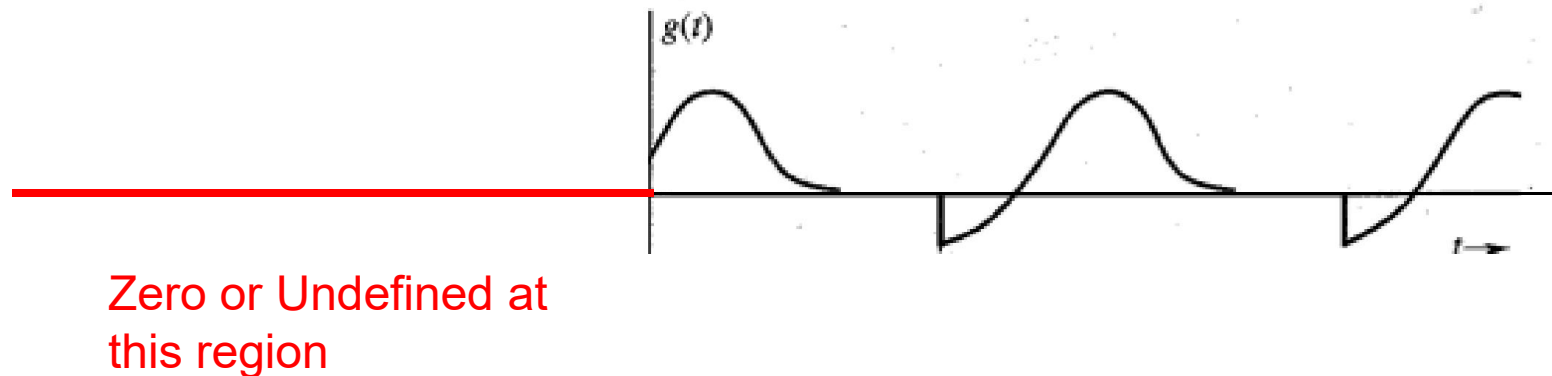
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Periodic and Aperiodic signal

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- cannot start at an finite time, say, $t = 0$, otherwise $g(t) = g(t + T_0)$ cannot be satisfied



At $t = -T_0$, $g(t) \neq g(t + T_0)$

Energy and Power signals

- Energy Signal: finite E_g

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt < \infty$$

- Power signal: finite P_g

$$0 < P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt < \infty$$

Energy and Power signals

$$E_g = \int_{-\infty}^{\infty} g^2(t) dt < \infty$$

$$0 < P_g = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} g^2(t) dt < \infty$$

- Finite E_g signal has zero P_g
- Finite P_g Signal has infinite E_g
- A signal **CANNOT** be both energy and power signal
- **Real life** signals are **energy** signals
- **Power signals** have **infinite duration**; impractical to generate
- **Periodic** signals are **power** signals

Deterministic and Random signals

- Deterministic
 - has complete physical description, mathematically or graphically
- Random
 - has only probabilistic description, e.g., mean value, rms, distribution
- All message signals are random

Signal Properties

Time shifting property

- Whatever happens in $g(t)$ at t second also happens in $\phi(t)$ T seconds later at instant $t + T$

$$\phi(t + T) = g(t)$$

Or,

$$\phi(t) = g(t - T)$$

Signal Properties

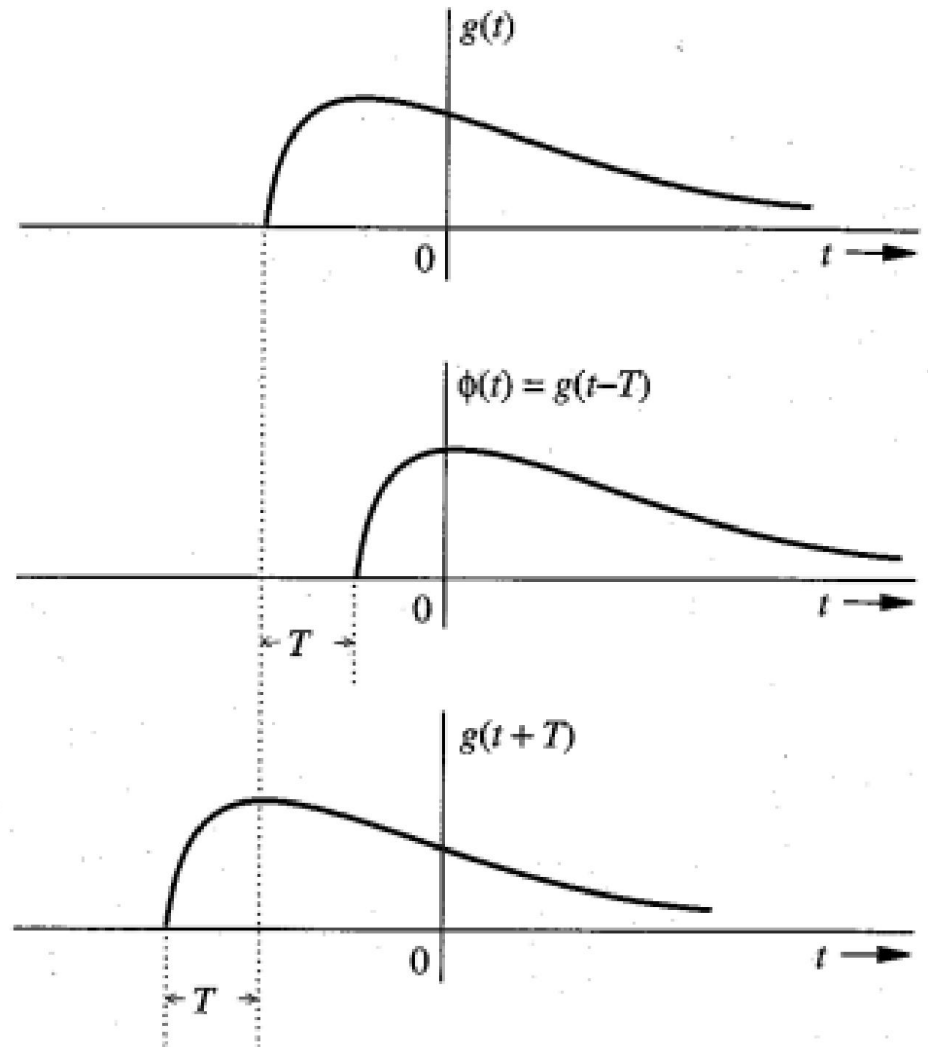
Time shifting property

$$\phi(t + T) = g(t)$$

$$\phi(t) = g(t - T)$$

Beginning T
seconds later

Beginning T
seconds earlier



Signal Properties

Time scaling property

- Compression or expansion
- Compression:
 - Whatever happens in $g(t)$ at t second also happens in $\phi(t)$ at t/a

$$\phi(t) = g(at), a > 1$$

Signal Properties

Time scaling property

- Compression or expansion
- Compression:
 - Whatever happens in $g(t)$ at t second also happens in $\phi(t)$ at t/a

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- Expansion:
 - Whatever happens in $g(t)$ at t second also happens in $\phi(t)$ at at

$$\phi(t) = g\left(\frac{t}{a}\right), a > 1$$

Signal Properties

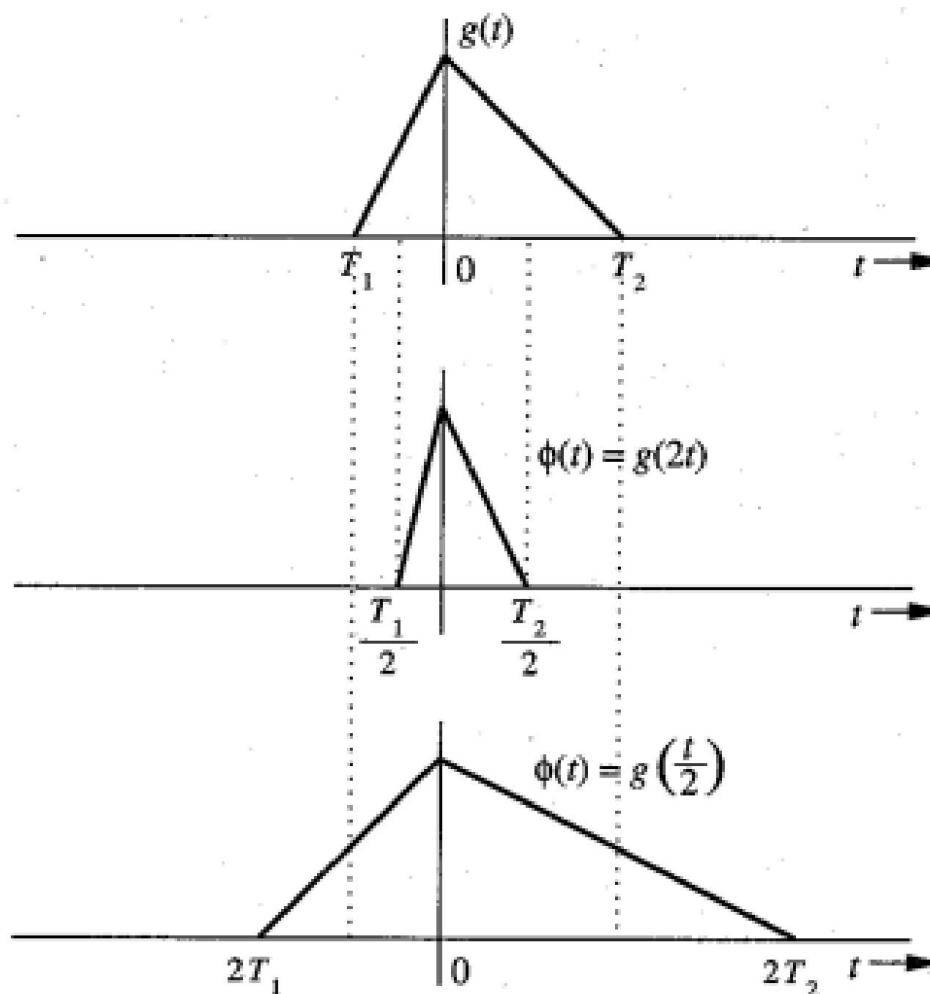
Time scaling property

$$\phi(t) = g(at)$$

$$\phi(t) = g\left(\frac{t}{a}\right)$$

Compression

Expansion



Signal Properties

Time inversion property

- Mirroring about vertical axis
- Whatever happens in $g(t)$ at t second also happens in $\phi(t)$ at $-t$
- Similar to time scaling where $a = -1$

$$\phi(-t) = g(t)$$

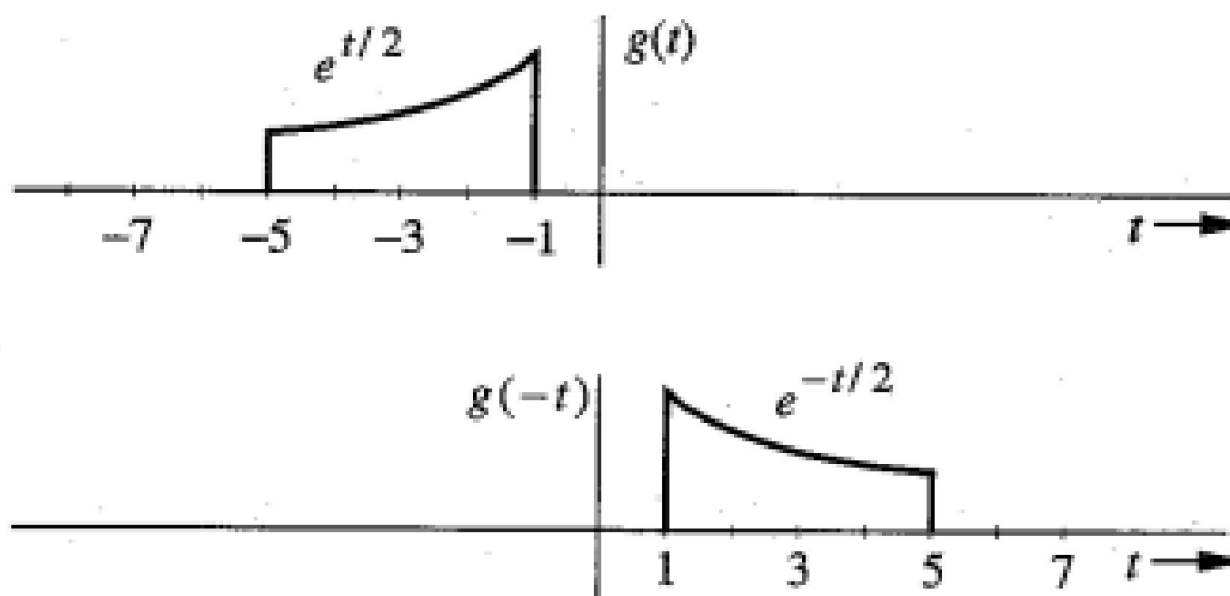
$$\phi(t) = g(-t)$$

Signal Properties

Example of Time inversion property

$$\phi(-t) = g(t)$$

$$\phi(t) = g(-t)$$



Unit Impulse Signal

- One of the most important signals, $\delta(t)$
- Also known as Dirac delta function

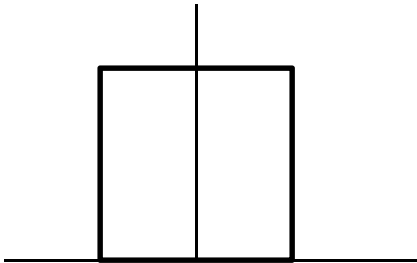
$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0 \end{cases}$$

with the constraint,

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

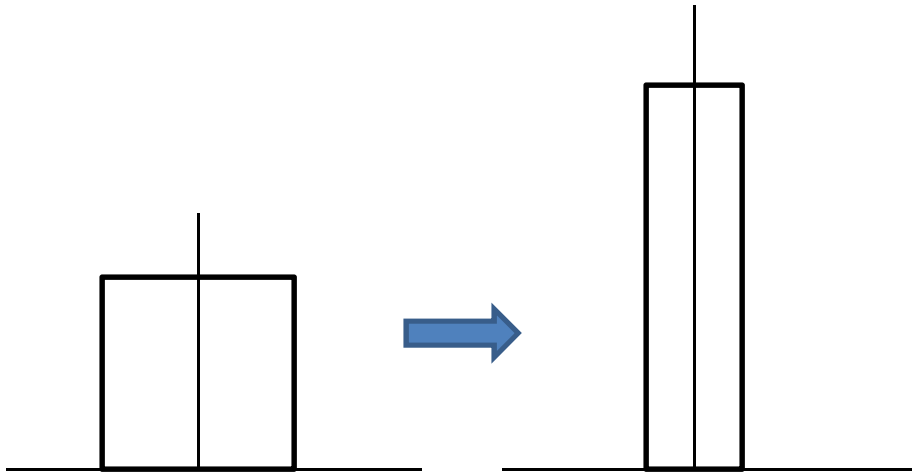
Unit Impulse Signal

A function with unit area under curve



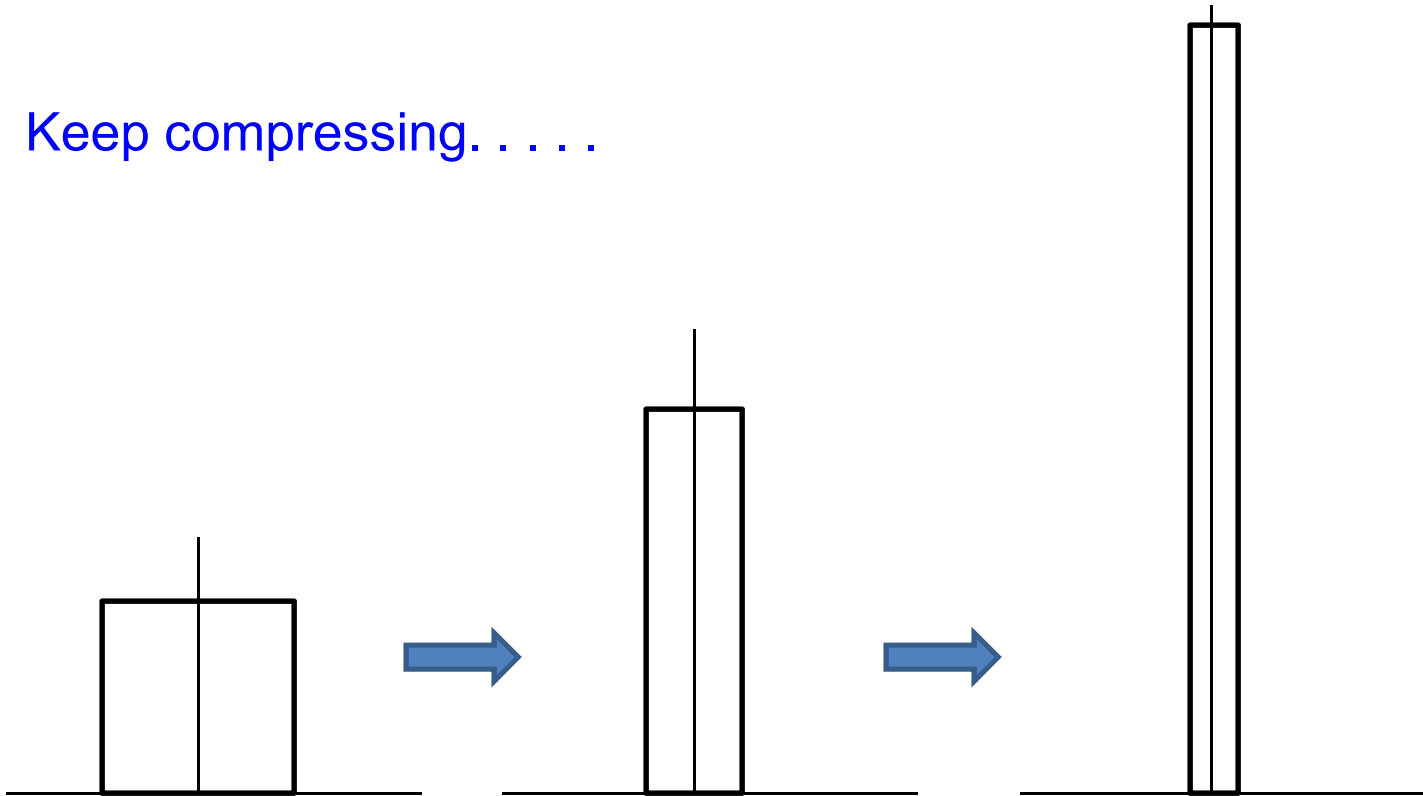
Unit Impulse Signal

Compress the function leaving
the area unchanged



Unit Impulse Signal

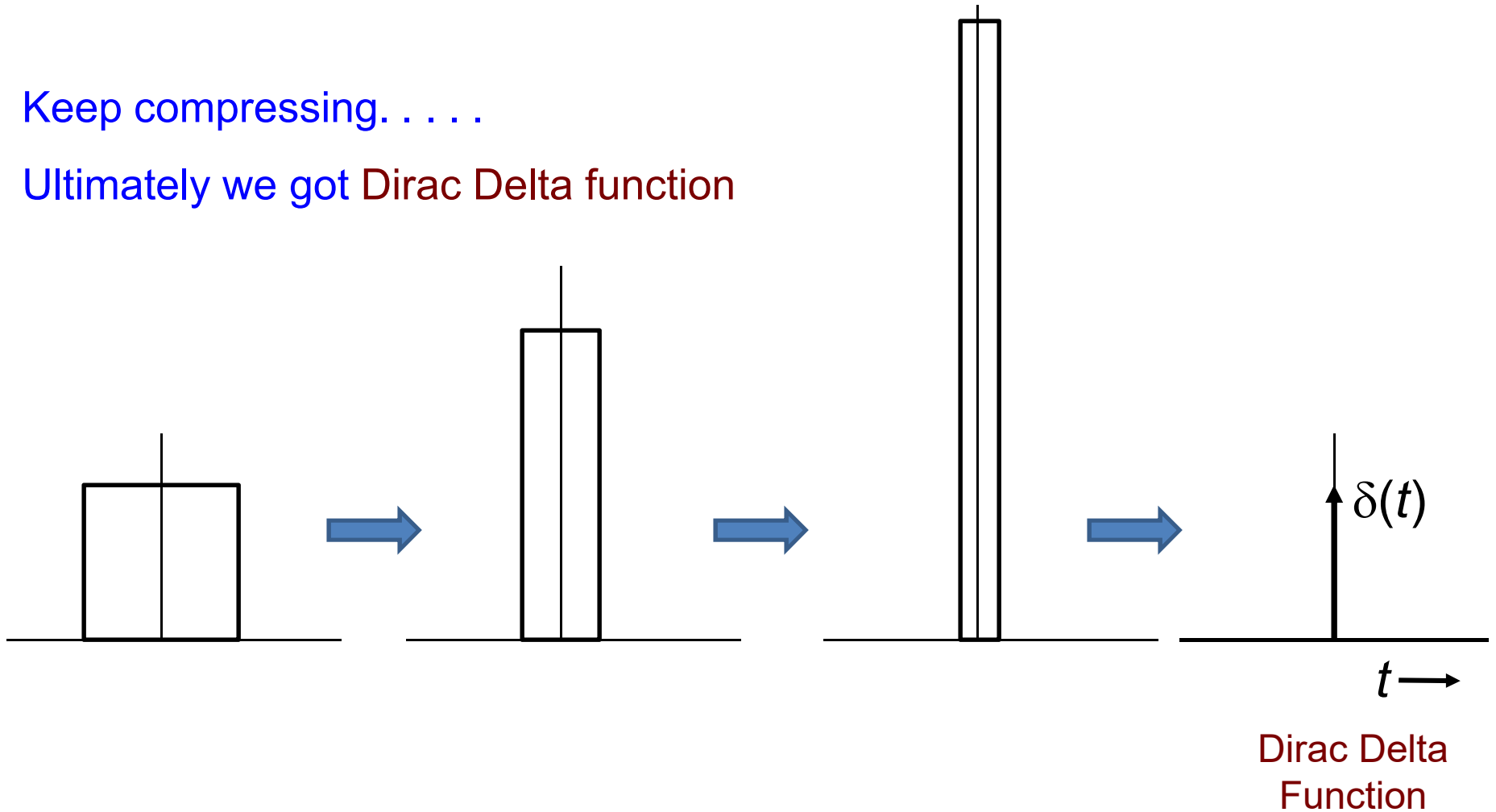
Keep compressing.



Unit Impulse Signal

Keep compressing.

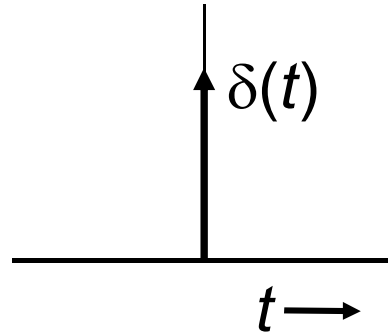
Ultimately we got Dirac Delta function



Unit Impulse Signal

$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0 \end{cases}$$

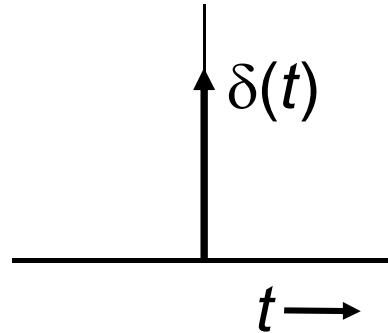
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Unit Impulse Signal

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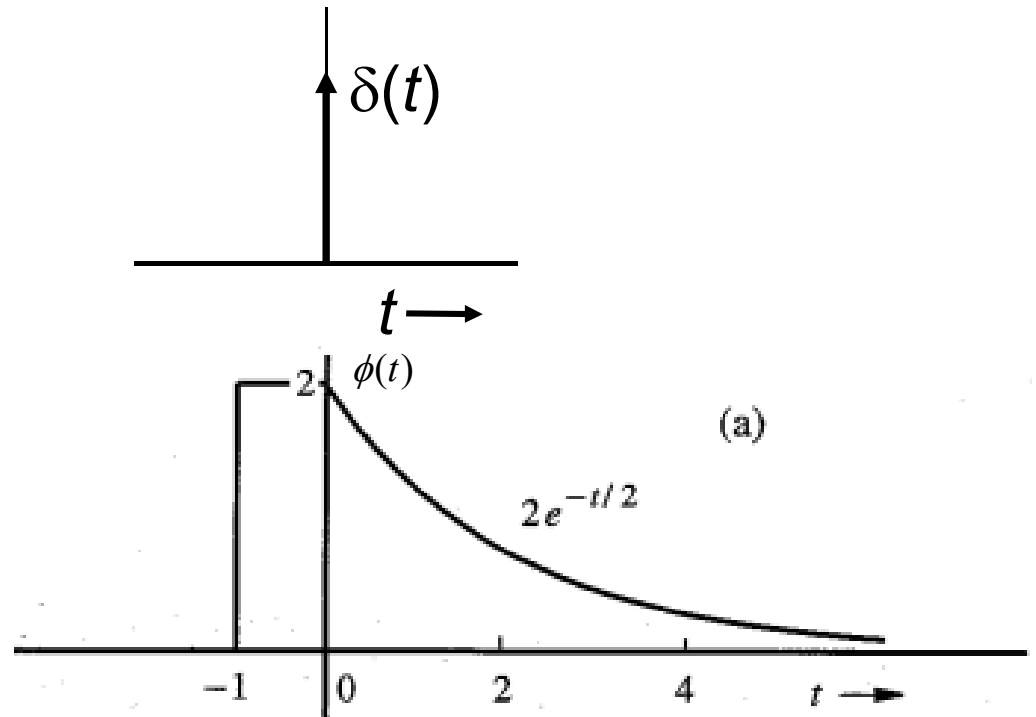
$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$



- Impulse location is at $t = 0$

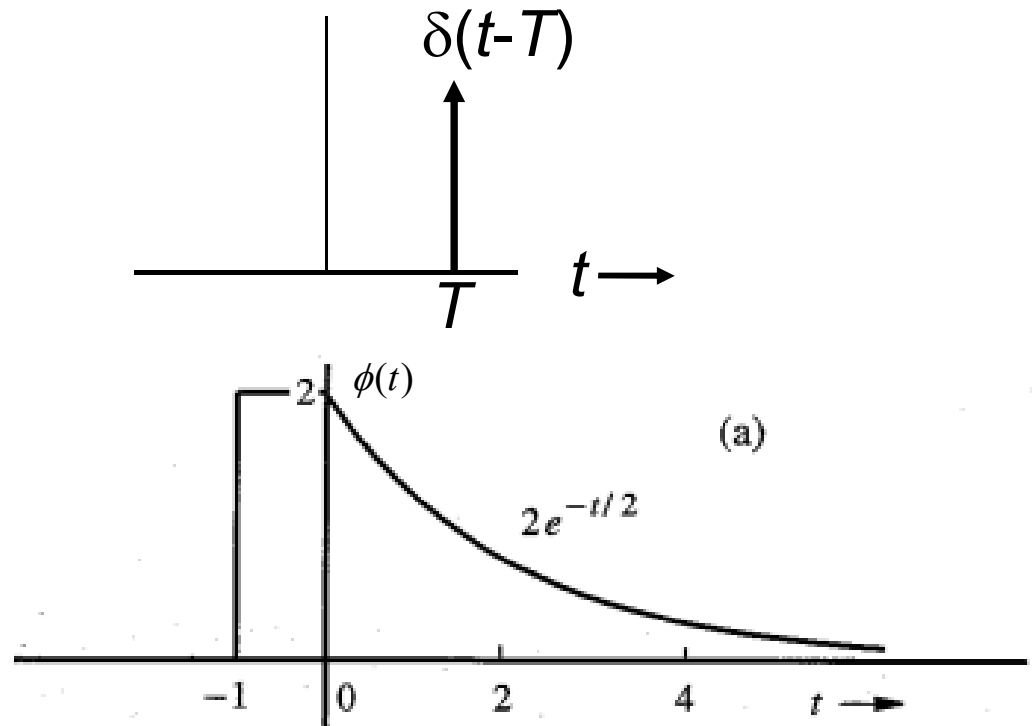
Multiplication of a Function by Impulse

$$\phi(t)\delta(t) = \phi(0)\delta(t)$$



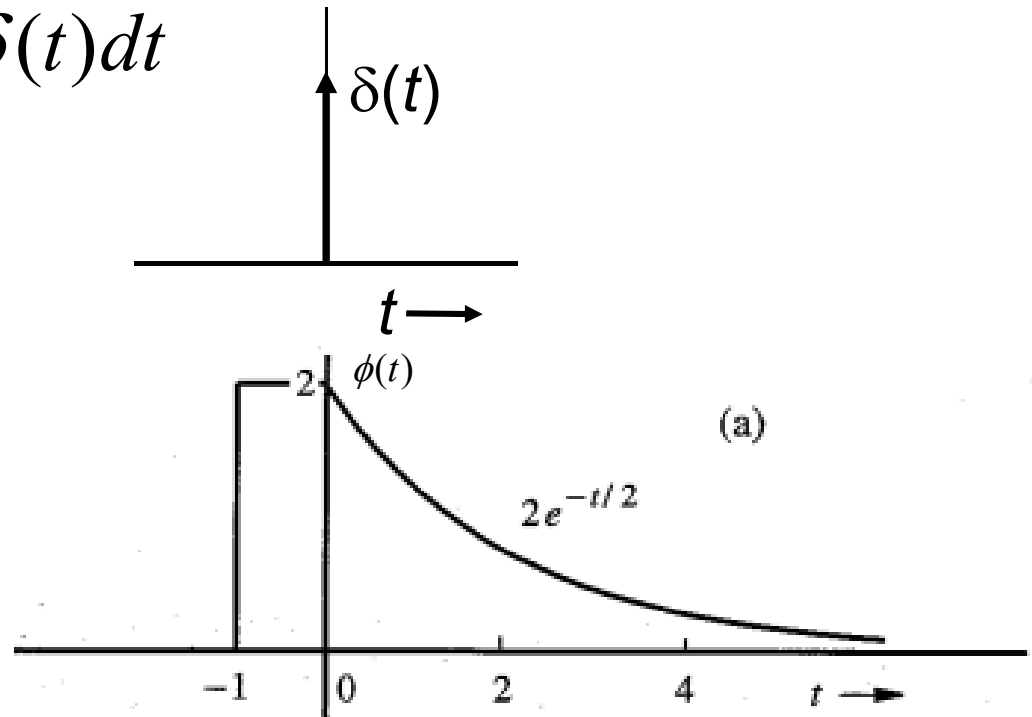
Multiplication of a Function by Impulse

$$\phi(t)\delta(t-T) = \phi(T)\delta(t-T)$$



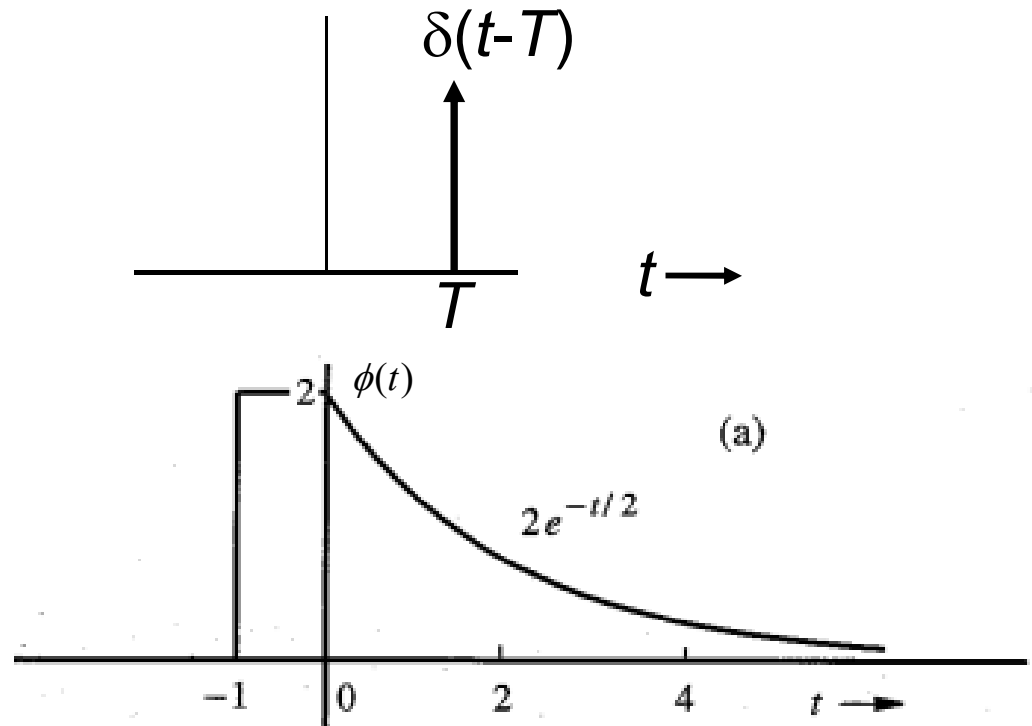
Sampling Property of the Unit Impulse Function

$$\int_{-\infty}^{\infty} \delta(t) \phi(t) dt = \phi(0) \int_{-\infty}^{\infty} \delta(t) dt \\ = \phi(0)$$



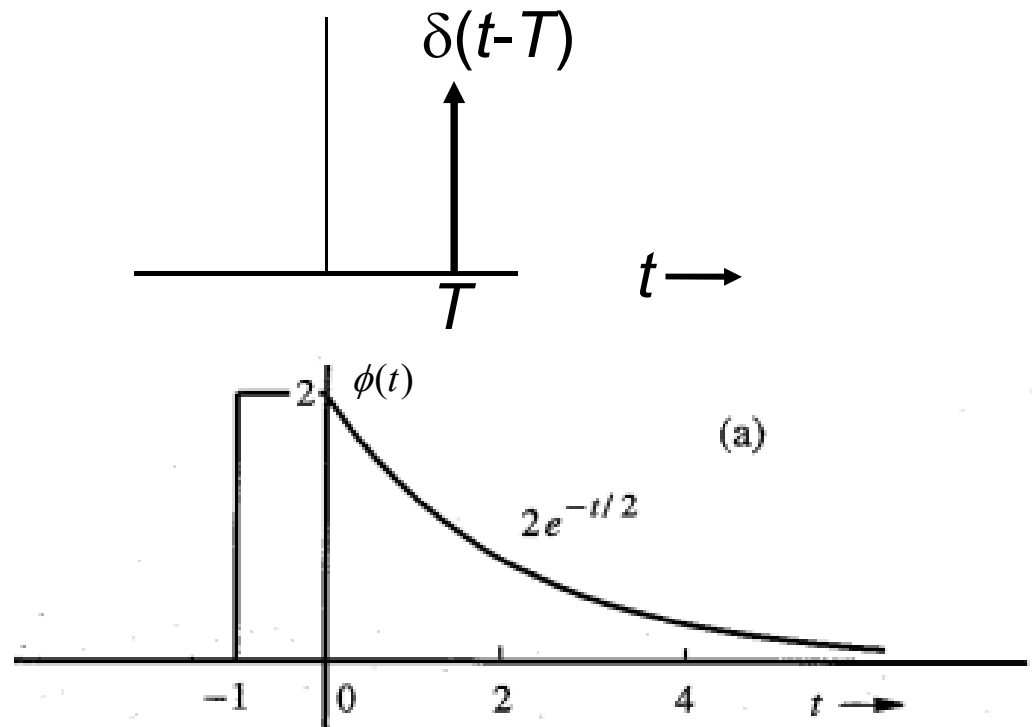
Sampling Property of the Unit Impulse Function

$$\begin{aligned} & \int_{-\infty}^{\infty} \delta(t-T)\phi(t)dt \\ &= \phi(T) \int_{-\infty}^{\infty} \delta(t-T)dt \\ &= \phi(T) \end{aligned}$$



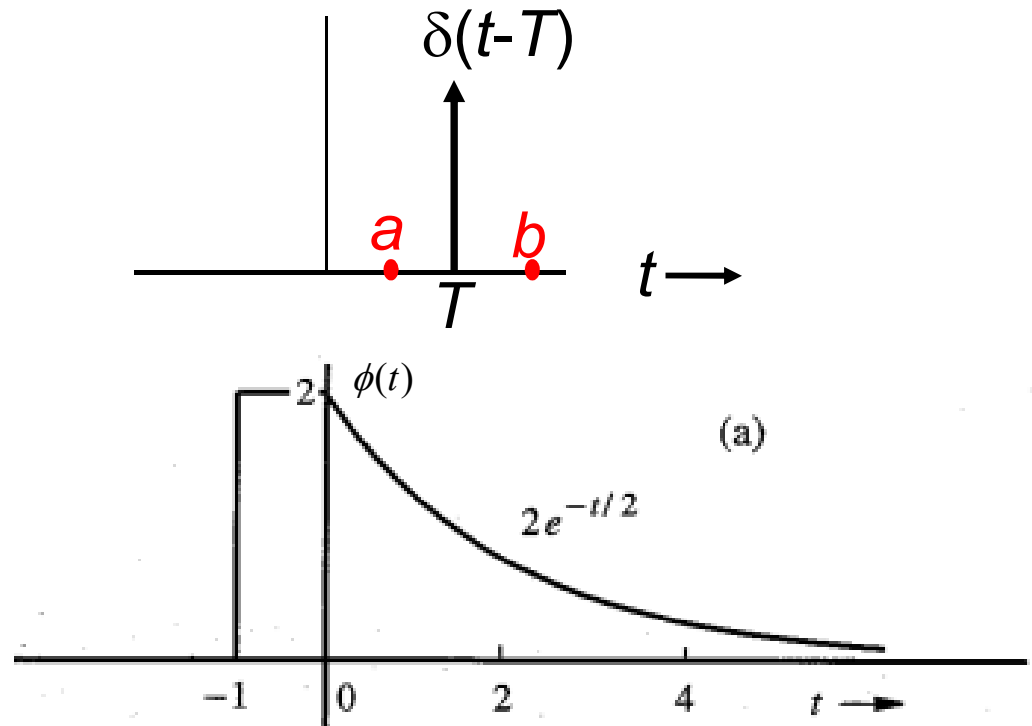
Sampling Property of the Unit Impulse Function

$$\int_a^b \delta(t-T)\phi(t)dt$$
$$= \phi(T) \int_a^b \delta(t-T)dt$$



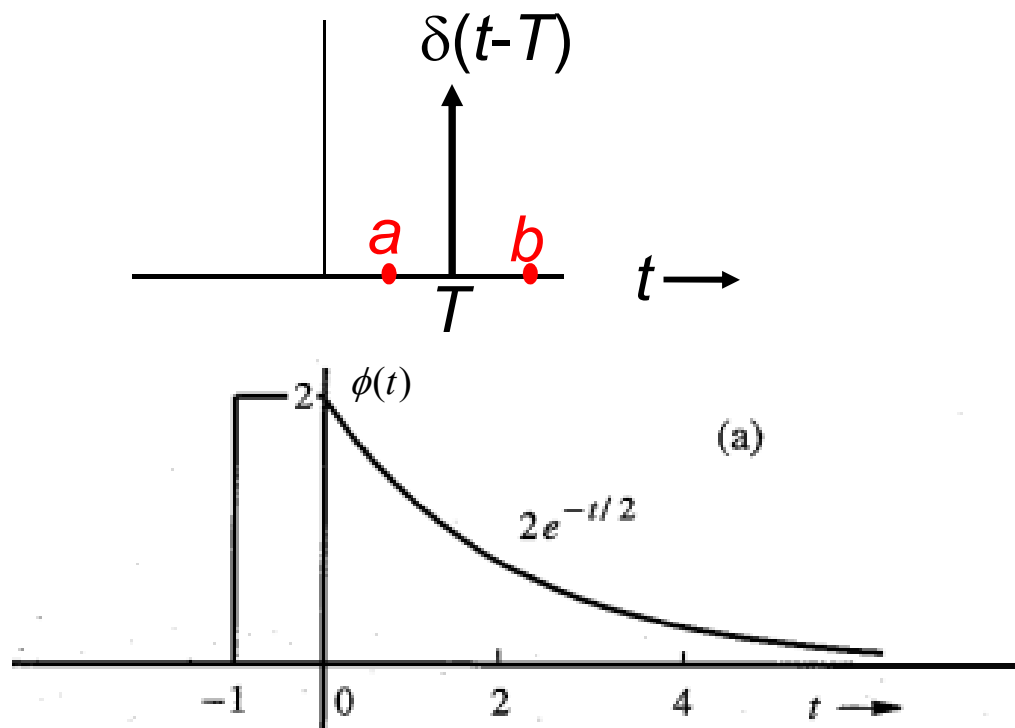
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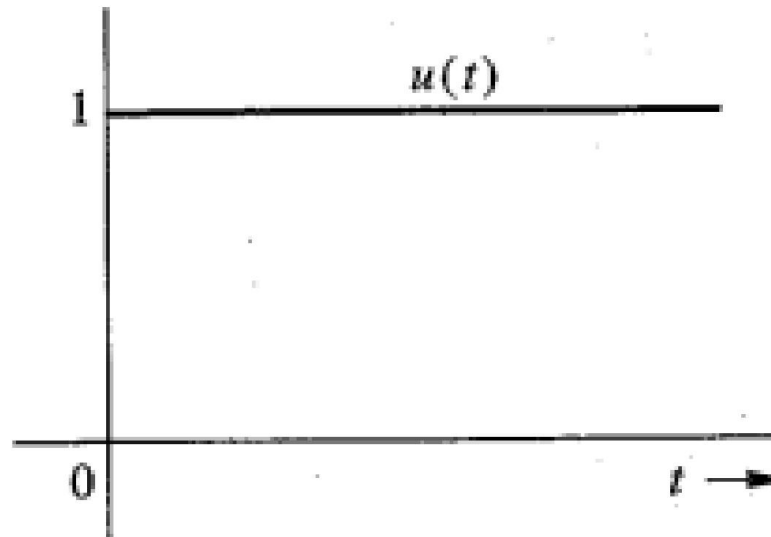


Sampling Property of the Unit Impulse Function

$$\begin{aligned} & \int_a^b \delta(t-T)\phi(t)dt \\ &= \phi(T) \int_a^b \delta(t-T)dt \\ &= \begin{cases} \phi(T) & a \leq T \leq b \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$



Unit step signal/function, $u(t)$



$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

Unit step signal/function, $u(t)$

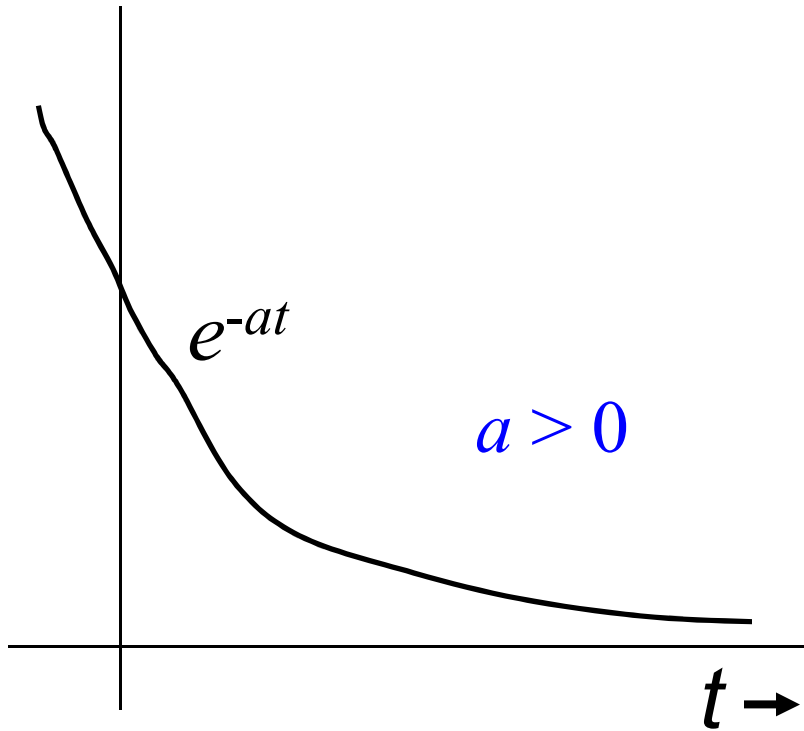
- Causal function, e.g., starts at $t = 0$

$$u(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases}$$

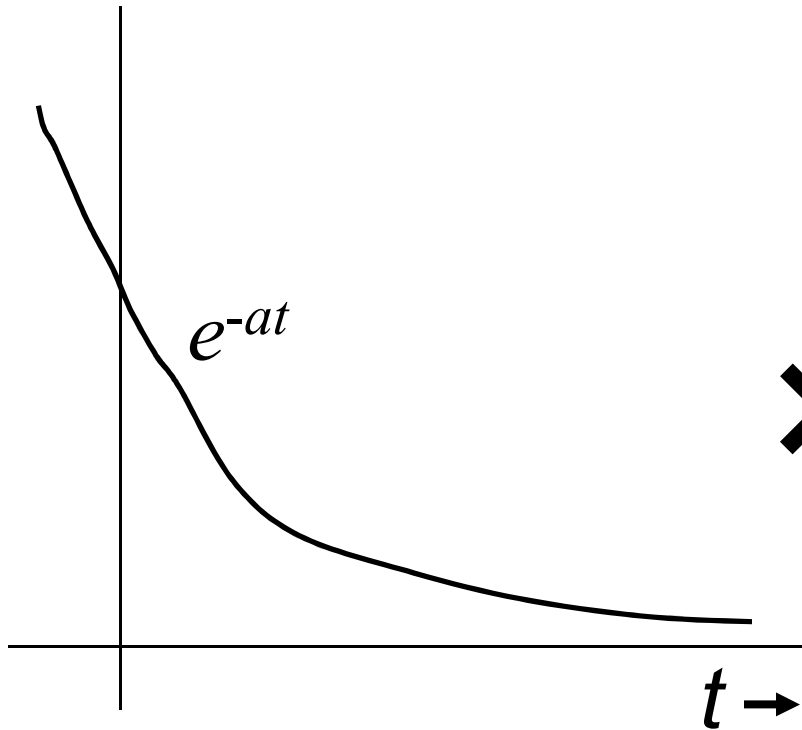
- $g(t)$ is causal if

$$g(t) = 0 \quad t < 0$$

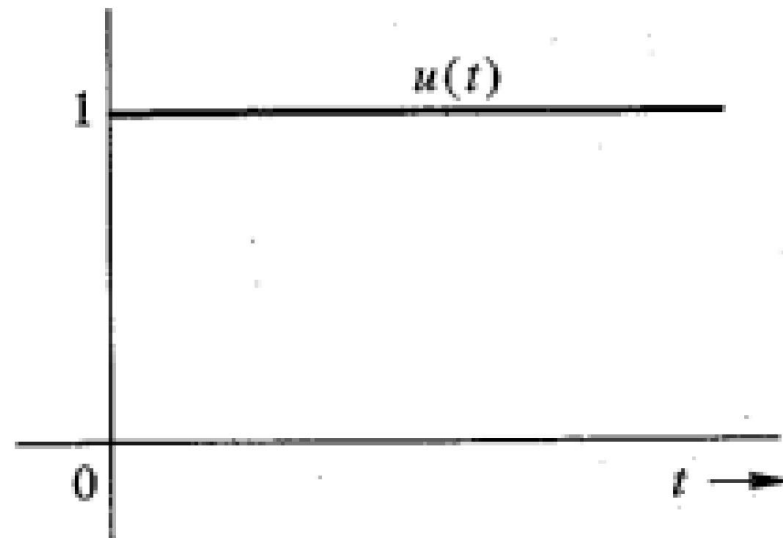
Use of $u(t)$ to start an everlasting function from $t = 0$



Use of $u(t)$ to start an everlasting function from $t = 0$



$\times$



Use of $u(t)$ to start an everlasting function from $t = 0$

